

Lecture 6.pdf

PDF

**Lecture 7.pdf**

PDF

AI Diagnostic Quiz Generator

You are creating a self-quiz for students taking lecture 6 and 7

Task:

I wanted to check how well I really understand the content of the lecture by prompting 5 conceptual questions from AI> I also wanted to understand any intuition I might missed after the lecture

Create 5 short conceptual questions that target the most important ideas in lecture 6 and 7

For each question, provide:

Answer (1–3 sentences)

Quick intuition (plain-language picture of what is happening)

Formal reasoning (definitions, equations, or logical steps)

At the end of each question, add an Assumption Check:

"⚠️ This question depends on assumptions not given in the prompt:
[list them]"

or

"✅ No missing assumptions required."

Rules:

Do NOT invent notation, theorems, or assumptions not in Section X.

If any part is uncertain due to missing context, explicitly say so.

Below is a **5-question diagnostic quiz** covering the core ideas from **Lecture 6 (RMS→RMS norm, maximal update parametrization)** and **Lecture 7 (Muon optimizer, spectral norm, Newton–Schulz)**, with answers, intuition, formal reasoning, and an explicit “assumption check.”

Everything strictly comes from the provided lecture images.

Citations: Lecture 6  Lecture 6 ; Lecture 7  Lecture 7

AI Diagnostic Quiz Generator — Lecture 6 & 7

Q1. What is the RMS→RMS induced matrix norm and why is it useful for choosing layer-scaled learning rates?

(Lecture 6 pp. 1–4)

Answer

It is a matrix norm defined by measuring both inputs and outputs in RMS units, giving

$$\|A\|_{\text{RMS} \rightarrow \text{RMS}} = \sqrt{\frac{d_{\text{in}}}{d_{\text{out}}}} \|A\|_2.$$

It is useful because it naturally rescales updates by layer width, giving layer-specific effective learning rates.

Quick intuition

You measure the *average* effect a matrix has on vectors whose entries have variance scaled by width. This cancels the effect of wider layers producing larger activations, so the optimizer treats different layers on comparable footing.

Formal reasoning

From Lecture 6 page 2,

$$\|A\|_{\alpha \rightarrow \beta} = \max_{\|x\|_{\alpha}=1} \|Ax\|_{\beta}$$

and choosing RMS–RMS norms gives (derived on page 2)

$$\|A\|_{\text{RMS} \rightarrow \text{RMS}} = \sqrt{\frac{d_{\text{in}}}{d_{\text{out}}}} \|A\|_2.$$

This scaling arises from the Xavier-initialization heuristic where activations are kept at $O(1)$ scale across layers.

Assumption Check

✓ No missing assumptions required.

Q2. Why does the RMS→RMS constraint $\|\Delta W\|_{\text{RMS} \rightarrow \text{RMS}} \leq \gamma$ imply a layer-dependent learning rate $\eta \propto 1/d_{\text{in}}$?

(Lecture 6 pp. 4, 11–13)

Answer

Because for rank-1 sign updates, $\|\Delta W\|_2 = \eta \|S\|_2$, and Lecture 6 shows $\|S\|_2 = \sqrt{d_{\text{in}} d_{\text{out}}}$. Enforcing the RMS→RMS bound forces $\eta \propto 1/d_{\text{in}}$.

Quick intuition

Rank-1 sign updates grow in magnitude with layer width. To keep updates equally “small” across layers, you must shrink the learning rate for wider layers.

Formal reasoning

From page 12 of Lecture 6:

- For rank-1 sign matrix S ,

$$\|S\|_2 = \sqrt{d_{\text{in}} d_{\text{out}}}.$$

- RMS→RMS constraint (also page 4):

$$\|\Delta W\|_{\text{RMS} \rightarrow \text{RMS}} = \eta \sqrt{\frac{d_{\text{in}}}{d_{\text{out}}}} \|S\|_2 \leq \gamma.$$

Substitute $\|S\|_2$:

$$\eta \sqrt{\frac{d_{\text{in}}}{d_{\text{out}}}} \sqrt{d_{\text{in}} d_{\text{out}}} = \eta d_{\text{in}} \leq \gamma \quad \Rightarrow \quad \eta \leq \gamma / d_{\text{in}}.$$

Assumption Check

- ✓ No missing assumptions required.

Q3. What problem does maximal-update parametrization (μP) solve, and how does layer-wise scaling emerge mathematically?

The explanation was correct and tied the width scaling to norm growth in a straightforward way. But the reasoning still feels a bit formula-dumpy; it cites $\|S\|^2 = d_{\text{in}} d_{\text{out}}$ without explaining where that comes from intuitively. Still, it did answer the specific question I had, and the intuition about rank-1 sign updates growing with width was helpful.

(Lecture 6 pp. 16–18)

Answer

μP solves the problem of transferring hyperparameters from small models to large ones by finding scaling rules under which training dynamics remain non-trivial as width grows. Layer-wise scaling emerges by ensuring activations and updates stay $O(1)$ as width increases.

Quick intuition

If layers get wider, activations and gradients naturally blow up or vanish. μP picks the “right units” so that the same learning rate schedule works across widths.

Formal reasoning

Lecture 6 (pages 16–18) shows a probabilistic analogy: only the normalization

$$\frac{1}{\sqrt{n}}(x_1 + \dots + x_n)$$

converges to a non-trivial limit.

To make neural network layers preserve $O(1)$ activations and $O(1)$ updates, we enforce (page 18):

$$\|h_\ell^2\|_{\text{RMS}} = \Theta(1), \quad \|\Delta W_\ell\|_{\text{RMS} \rightarrow \text{RMS}} = \Theta(1).$$

These yield the scaling $\eta \sim 1/d_{\text{in}}$, giving consistent behavior across widths.

Assumption Check

⚠️ **Depends on the assumption that nonlinearities and biases are ignored** (noted on page 19).

No additional assumptions beyond the lecture.

This one was actually one of the stronger explanations. It frames μP as finding the “non-trivial limit” as width grows, which matches how I think about it. But it still glosses over some things — for example, it says “activations and updates stay $\Theta(1)$ ” without explaining the probabilistic mechanics behind that. The assumption flagging was good though; I appreciated it admitting that dropping nonlinearities is a big caveat.

Q4. Why does the Muon optimizer aim to approximate the semi-orthogonal update $UV^\top$ rather than compute the full SVD $U\Sigma V^\top$?

(Lecture 7 pp. 2–6)

Answer

Because the spectral-norm-constrained optimizer's steepest descent direction is $UV^\top$, but computing it exactly requires an SVD. Muon replaces Σ with the identity, giving a direction that is "good enough" while being far cheaper.

Quick intuition

Muon wants to move in the direction that treats all directions uniformly (not dominated by a large singular value). Instead of computing the exact singular values, it strips them away and keeps only orientation.

Formal reasoning

Lecture 7 (page 5) shows the spectral-norm optimization:

$$\arg \min_{\|\Delta W\|_2 \leq \eta} \langle \nabla W, \Delta W \rangle = \eta UV^\top.$$

But page 6 notes the SVD is expensive. Muon uses the Newton–Schulz iteration applied to $GG^\top$ to approximate a matrix with singular values all ≈ 1 , so that

$$U\Sigma V^\top \approx UV^\top.$$

Assumption Check

✓ No missing assumptions required.

Q5. What is the role of the Newton–Schulz iteration in Muon, and why do polynomials that commute with the SVD matter?

(Lecture 7 pp. 7–10)

Answer

Newton–Schulz iteratively pushes singular values toward 1, producing a matrix whose orientation matches the gradient but is “orthogonalized.” Polynomials that commute with the SVD preserve singular vectors, so the iteration modifies only singular values.

Quick intuition

The iteration repeatedly “reshapes” the matrix so its stretching becomes uniform in all directions. You keep the direction but remove anisotropy.

Formal reasoning

Lecture 7 page 7 shows that if

$$p(X) = a_0 X + a_1 X(X^\top X) + \dots,$$

then

$$p(U\Sigma V^\top) = U p(\Sigma) V^\top.$$

Thus singular vectors stay the same.

Page 8–9 shows the specific polynomial

$$p(x) = \frac{3}{2}x - \frac{1}{2}x^3,$$

which maps $x \mapsto 1$ for $x > 0$. Iterating p (Newton–Schulz) pushes all singular values toward 1, giving the Muon update direction.

This explanation was helpful in the sense that it clarified: “the polynomial changes singular values but keeps singular vectors.” That’s basically what I needed.

Assumption Check

⚠️ Depends on assuming singular values are first scaled to lie in $(0, 1)$ (explicitly stated on Lecture 7 page 9).

If you'd like, I can generate a **multiple-choice version**, a **homework-style worksheet**, or an **Ed-post-ready prompt** for students.

Sources

create computational questions that are mapped to these core concepts

Below are **computational questions** (not purely conceptual) that directly map to the **five core concepts** you identified from Lecture 6 & 7.

Each question requires a calculation, small derivation, or numerical check.

If you want, I can also generate **solutions**, **rubrics**, or **multiple difficulty tiers**.

Citations: Lecture 6  Lecture 6 ; Lecture 7  Lecture 7

Computational Questions Aligned to Core Concepts

Q1 — Compute an RMS→RMS induced matrix norm*(Maps to: RMS→RMS norm formula from Lecture 6 pages 1–4)*You are given a matrix $A \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$ with

$$d_{\text{in}} = 8, \quad d_{\text{out}} = 2, \quad \|A\|_2 = 5.$$

Compute the RMS→RMS induced norm

$$\|A\|_{\text{RMS} \rightarrow \text{RMS}}.$$

Expected work: plug into

$$\|A\|_{\text{RMS} \rightarrow \text{RMS}} = \sqrt{\frac{d_{\text{in}}}{d_{\text{out}}}} \|A\|_2.$$

✓ No missing assumptions.

Q2 — Compute the layer-wise learning rate that satisfies the RMS→RMS update constraint*(Maps to: learning rate scaling $\eta \propto 1/d_{\text{in}}$, Lecture 6 pages 11–13)*

Suppose your update uses sign-SGD with

$$\Delta W = -\eta S,$$

where S is rank-1 with

$$\|S\|_2 = \sqrt{d_{\text{in}} d_{\text{out}}}.$$

GPT doesn't
fully walk
through the
computation
— it just
states the
formula
again. It's
still a fine
question.

Let

$$d_{\text{in}} = 512, \quad d_{\text{out}} = 1024, \quad \gamma = 0.001,$$

and impose the constraint

$$\|\Delta W\|_{\text{RMS} \rightarrow \text{RMS}} \leq \gamma.$$

Compute the **maximum allowable** η .

Expected work: enforce

$$\eta d_{\text{in}} \leq \gamma.$$

✓ No missing assumptions.

Q3 — Check whether a given NN layer satisfies the μP feature-learning conditions

(Maps to: conditions on $\|h_{\ell}^2\|_{\text{RMS}}$ and $\|\Delta W_{\ell}\|_{\text{RMS} \rightarrow \text{RMS}}$, Lecture 6 pages 18–20)

A layer has the following empirical measurements during training:

- Hidden activations:

$$\|h_{\ell}^2\|_{\text{RMS}} = 0.95.$$

- Weight update under RMS→RMS norm:

$$\|\Delta W_{\ell}\|_{\text{RMS} \rightarrow \text{RMS}} = 0.02.$$

Suppose the model width is scaled by a factor of $k = 4$.

Assuming you keep the same learning rate, the new update norm is observed to be:

The question is solid. It forces me to compute and compare norms across widths, which is useful. But the AI didn't walk through any numbers — it left the entire burden on me, which is fine but slightly abrupt.

$$\|\Delta W_\ell\|_{\text{RMS} \rightarrow \text{RMS}_{\text{new}}} = 0.08.$$

Question:

Does this violate the μP desideratum that

$$\|\Delta W_\ell\|_{\text{RMS} \rightarrow \text{RMS}} = \Theta(1)$$

as width grows? Show your reasoning numerically.

⚠ Depends on the assumption (Lecture 6 page 19) that nonlinearity and bias can be ignored for the scaling argument.

Q4 — Compute the spectral-norm–constrained steepest-descent update direction

(Maps to spectral-norm optimization and semi-orthogonal update on Lecture 7 pages 2–5)

Given the gradient matrix

$$G = U \Sigma V^\top \quad \text{with} \quad \Sigma = \text{diag}(9, 3, 1),$$

and learning rate $\eta = 0.01$,

Compute the steepest-descent update under the spectral norm constraint:

$$\Delta W^* = \arg \min_{\|\Delta W\|_2 \leq \eta} \langle G, \Delta W \rangle.$$

Expected work: use the formula

$$\Delta W^* = \eta U V^\top.$$

✓ No missing assumptions.

Q5 — Apply one Newton–Schulz iteration to modify singular values

(Maps to Newton–Schulz iteration and polynomial $p(x) = \frac{3}{2}x - \frac{1}{2}x^3$, Lecture 7 pp. 7–9)

Let a singular value be

$$\sigma = 0.4.$$

Apply the polynomial

$$p(x) = \frac{3}{2}x - \frac{1}{2}x^3.$$

Compute the updated singular value $p(\sigma)$.

Then compute the second iteration $p(p(\sigma))$.

Goal: show numerically that the singular value moves closer to 1.

⚠ Depends on the Lecture 7 assumption that singular values have been normalized into $(0, 1)$ beforehand (page 9).

If you want, I can now generate **full worked solutions**, a **PDF worksheet**, or an **auto-graded version** compatible with Gradescope / Canvas / Ed.

Sources

This question is great for intuition — it's exactly the sort of “plug a number in and see it move toward 1” demonstration I needed. The assumption caveat was also correct and worth noting.

link these techniques to classic problems

Below are **classic problems in numerical optimization and machine learning** that map naturally to the *same mathematical structures* underlying **RMS→RMS norms, maximal update parametrization, spectral-norm–constrained updates, and Newton–Schulz–orthogonalization** from Lecture 6 and 7. Each section shows **(1) what classic problem it mirrors, (2) how the lecture technique resolves the difficulty, and (3) a concrete example computational question** you could assign.

Citations: Lecture 6  Lecture 6 ; Lecture 7  Lecture 7

In the end I wanted to see if it is able to link lecture concepts to classical problems as an extension

1. RMS→RMS Norm → Classic Problem: Conditioning in Least Squares / Linear Regression

Classic problem

When solving

$$\min_x \|Ax - b\|_2^2,$$

the conditioning of A determines stability. If columns of A have wildly different scales, gradients propagate unevenly across coordinates.

How the lecture technique links

RMS→RMS norm (Lecture 6 pages 1–3) rescales the action of a matrix by width, similar to *column normalization* in regression.

It provides a meaningful unit: "average effect per coordinate."

This is exactly what one does in classic LS problems via standardization.

Classic computational question

Let A have columns with variance scaling like $1/d_{\text{in}}$. Compute the RMS→RMS norm and show that using it is equivalent to preconditioning the regression problem so that all coordinates have comparable influence.

2. Learning-Rate Scaling ($\eta \propto 1/d_{\text{in}}$) → Classic Problem: Gradient Descent on Quadratic Bowls

Classic problem

In minimizing a quadratic

$$f(w) = \frac{1}{2} w^\top Q w,$$

the stable step size satisfies $\eta \leq 2/\lambda_{\max}(Q)$.

If dimension grows and eigenvalues grow with it, the stable learning rate shrinks.

How the lecture technique links

Lecture 6 (pp. 11–13) shows

$$\eta_{\max} \propto 1/d_{\text{in}}.$$

This mirrors the quadratic-bowl fact that as width increases, curvature increases, so step size must decrease.

Classic computational question

Consider $Q = I_d + \mathbf{1}\mathbf{1}^\top$. Compute $\lambda_{\max}(Q)$ and show that the stable GD step-size scales like $1/d$. Compare this with the RMS→RMS-derived rule $\eta \propto 1/d_{\text{in}}$.

3. μ P Scaling → Classic Problem: CLT Scaling in Classical Probability

Classic problem

To get a non-degenerate limit of

$$S_n = x_1 + \cdots + x_n,$$

you must scale by $1/\sqrt{n}$ (Lecture 6 page 16).

Any other scaling gives trivial behavior.

How the lecture technique links

μ P (Lecture 6 pages 16–18) is **literally the NN version of “choose the only scaling that keeps the limit non-trivial.”**

Instead of sums of random variables, we have transformations of activations and gradients as width grows.

Classic computational question

Derive the variance of the pre-activation Wh when W has entries of variance $1/d_{\text{in}}$. Show that without Xavier-style/ μ P scaling, activations blow up or vanish as width grows.

4. Spectral-Norm–Constrained Updates → Classic Problem: Trust-Region Methods

Classic problem

Trust-region updates solve

$$\min_{\|p\| \leq \Delta} \nabla f^\top p + \frac{1}{2} p^\top H p.$$

When the quadratic term is ignored (or very small), the direction reduces to the steepest descent under a norm constraint.

How the lecture technique links

Lecture 7 pages 2–3 show the analogous neural-net update:

$$\min_{\|\Delta W\|_2 \leq \eta} \langle G, \Delta W \rangle \Rightarrow \Delta W = \eta U V^\top.$$

This is the trust-region step under a spectral norm ball.

Classic computational question

For a quadratic model with gradient g , compute the trust-region step under a 2-norm constraint and verify that the solution is $-\Delta g / \|g\|_2$. Then extend the analogy to matrices: compute $\eta U V^\top$ for a given SVD.

5. Newton–Schulz Orthogonalization → Classic Problems: Iterative Matrix Inversion & Polar Decomposition

Classic problem A: Matrix inverse via Newton iteration

Given A^{-1} , Newton's method for inversion uses

$$X_{k+1} = X_k(2I - AX_k).$$

Classic problem B: Polar decomposition

To compute the unitary factor U of

$$A = UH,$$

Newton iterations shrink the singular values of A toward 1, leaving singular vectors unchanged.

How the lecture technique links

Lecture 7 pages 7–10 show Newton–Schulz iterations:

- Apply a polynomial $p(X) = \frac{3}{2}X - \frac{1}{2}X^3$.
- Because p commutes with SVD, it modifies singular values but preserves orientation.
This is *exactly the same mathematics* as polar decomposition algorithms.

Classic computational question

Given a matrix $A = \begin{pmatrix} 2 & 0 \\ 0 & 0.5 \end{pmatrix}$, apply one polar-decomposition Newton step:

$$A_{k+1} = \frac{1}{2}A(3I - A^T A).$$

Compute the new singular values and show they move closer to 1, exactly as in Muon's Newton–Schulz updates.

Overall the
link is valid
but lacks
mathematical
explanation

6. Rank-1 Gradients in Mini-batch Size 1 → Classic Problem: Low-Rank Approximations / Power Method

Classic problem

When a gradient is rank-1, it can be written as

$$G = uv^\top$$

and its spectral norm is simply $\|u\|\|v\|$.

This is the same idea behind the *power method*—rank-1 structure identifies dominant singular directions.

How the lecture technique links

Lecture 6 pages 9–12 show that sign-SGD on mini-batch size 1 produces a rank-1 update, making the spectral norm equal to the Frobenius norm.

This is analogous to the *best rank-1 approximation* problem.

Classic computational question

Given $G = uv^\top$ with $u = [1, 2, 2]$ and $v = [3, 2]$, compute $\|G\|_2$, $\|G\|_F$, and verify that they coincide as predicted for rank-1 matrices.

7. Feature-Learning Conditions (Stability of Activations) → Classic Problem: Stability of Iterative Dynamical Systems

Classic problem

For the recurrence

$$x_{k+1} = Ax_k,$$

bounded solutions require $\|A\| < 1$.

Otherwise the system explodes.

How the lecture technique links

Lecture 6 pages 18–20 require

$$\|h_\ell^2\|_{\text{RMS}} = \Theta(1)$$

and

$$\|\Delta W_\ell\|_{\text{RMS} \rightarrow \text{RMS}} = \Theta(1)$$

to prevent exploding or vanishing activations during width scaling.

This is exactly a *stability condition* for a high-dimensional dynamical system.

Classic computational question

Given a linear layer with weight variance scaled incorrectly (e.g., variance = 1 instead of $1/d_{\text{in}}$), compute the expected RMS of activations across depth and show that they blow up exponentially.

If you want, I can now produce:

- ✓ A full assignment set combining all computational + conceptual + classic-problem links
- ✓ A lecture-7-only version focused on Muon and orthogonalization
- ✓ A table that maps “lecture idea → classical analogy → computational exercise”
- ✓ A PDF worksheet formatted for students

Just tell me the style you want.